

LaTeX Code for Probability Lecture Notes

Task Instructions: Using the same structure as the example provided above, generate a comprehensive lecture reference material for a reading-based/closed-notes exam.

Requirements:

Faithfulness & Detail: Include all definitions, notation, and assumptions exactly as presented. Every formula,

Step-by-Step Explanations: For all proofs and derivations, provide the complete step-by-step logic shown in

Scope: Stick strictly to the provided PDF. Do not invent new content or add external "intuition" not present in

Correctness & Tone: Maintain a formal academic tone. Prioritize mathematical precision and clarity in the lay

Structure: Adhere strictly to the organizational pattern (headings, lists, and layout) demonstrated in the "Exa



Lecture Scribe: Joint Random Variables and Distributions

Course: CSE400 – Fundamentals of Probability in Computing

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1. Introduction to Joint Random Variables

In scientific analysis and real-world applications, it is often necessary to observe and analyze two or more random variable

Real-World Examples:

- **Smart Transportation:** X = Vehicle Speed, Y = Traffic Density.
- **Wireless Networks:** X = Signal-to-Noise Ratio (SNR), Y = Data Throughput.
- **Logistics:** X = Wind Speed, Y = Delivery Time for drone delivery.
- **General Biology/Social Science:** Height and weight of giraffes, IQ and birthweight of children, or exercise frequency

Key Concepts:

- **Joint Distribution:** Allows the computation of probabilities for events involving multiple variables and reveals their de
 - **Independence:** The simplest case where variables do not influence each other.
 - **Correlation/Covariance:** Measures used to quantify the nature of dependence when variables are not independent.
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2. Discrete Case: Joint Probability Mass Function (PMF)

Setup and Definition

Suppose X and Y are discrete random variables with possible values $X \in \{x_1, x_2, \dots, x_n\}$ and $Y \in \{y_1, y_2, \dots, y_m\}$.

Definition: The Joint PMF is defined as:

This represents the probability that the joint outcome occurs simultaneously.

Key Properties:

- 1. **Non-negativity:** $0 \leq p(x_i, y_j) \leq 1$ for all i, j .
- 2. **Normalization:** The sum of all joint probabilities must equal 1:

Table Representation:

Joint PMFs are typically organized in a table where rows represent values of X and columns represent values of Y .

3. Worked Example: Rolling Two Dice

Experiment: Roll two fair dice.

Let:

- X = Value on the first die.
- T = Total (sum) of both dice.

Construction:

Since each outcome (i, j) of two dice has a probability of $1/36$, we construct the joint probability table for (X, T) .

$X \backslash T$	2	3	4	5
1	1/36	1/36	1/36	1/36
2	0	1/36	1/36	1/36
3	0	0	1/36	1/36
4	0	0	0	1/36
5	0	0	0	0
6	0	0	0	0

Observations:

- Entries are either 0 or $1/36$.
- X and T are NOT independent because the possible values of T depend on the value of X .

4. Continuous Case: Joint Probability Density Function (PDF)

The continuous case is analogous to the discrete case, with the following transitions:

- Discrete values $\rightarrow$ Continuous intervals.
- Joint PMF $\rightarrow$ Joint PDF.
- Sums $\rightarrow$ Double integrals.

Setup and Definition

If $X \in [a, b]$ and $Y \in [c, d]$, the pair (X, Y) takes values in the product region $[a, b] \times [c, d]$.

Definition: A function $f(x, y)$ is the joint PDF of X and Y if:

$$P((X$$

Key Properties:

1. **Non-negativity:** $f(x, y) \geq 0$.
2. **Normalization:** The total volume under the surface must be 1:
3. **Density vs. Probability:** The joint PDF $f(x, y)$ can take values **greater than 1** because it is a probability density, not

5. Comprehensive Example: Continuous Joint PDF

Given: $f(x, y) = 4xy$ for $0 \leq x \leq 1$ and $0 \leq y \leq 1$.

Task:

1. Show $f(x, y)$ is a valid joint PDF.
2. Compute $P(A)$ for $A = \{X < 0.5, Y > 0.5\}$.

Step-by-Step Solution:

1. Verification of Validity:

Check if the double integral over the unit square $[0, 1]^2$ equals 1.

$$\int_0^1$$

The function is non-negative and integrates to 1; thus, it is valid.

2. Computing Probability $P(A)$:

Integrate the PDF over the region $0 < x < 0.5$ and $0.5 < y < 1$.

Step 1: Inner integral with respect to x :

Step 2: Outer integral with respect to y :

$$\int_{0.5}^1 0.$$

Result: $P(A) = 0.1875$.

6. Joint Cumulative Distribution Function (Joint CDF)

The Joint CDF provides the probability that X is less than or equal to x AND Y is less than or equal to y .

Definition:

Calculation Methods:

- **Continuous Case:**

- **Discrete Case:**

Properties of Joint CDF:

1. **Monotonicity:** $F(x, y)$ is non-decreasing in both x and y .
2. **Boundary Conditions:**
 - $F(-\infty, y) = F(x, -\infty) = 0$.
 - $F(\infty, \infty) = 1$.
3. **Relationship to PDF:** The joint PDF is the mixed partial derivative of the joint CDF: